

Analyses: Feature Value Distributions

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Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(stringr)
library(ggplot2)
library(ggrepel)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggExtra)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'

## The following object is masked from 'package:dplyr':
##
##   combine

library(ggpubr)
```

Load Data

Load csv file with quantitative feature values for sign sequences per object (paleolithic signs) and tablet (cuneiform). Comparative sample: A comparative sample of feature values for languages across the world is derived from the Text Data Diversity (TeDDi) sample (Moran et al 2022).

```
estimations.df <- read.csv("~/Github/UpperPalCombinatorics/results/features_combined.csv")
head(estimations.df)
```

```
##      subcorpus      sequence num.char huni.chars hrate.chars  ttr.chars
## 1 Aurignacien ||| //||| _ V      12  1.8962406   1.5295563  0.4166667
## 2 Aurignacien      vvvvv        5  0.0000000   0.5711903  0.2000000
## 3 Aurignacien      vvvv vvvv     9  0.5032583   0.9060047  0.2222222
## 4 Aurignacien vvvvvvvvvvvvvvvv  14  0.0000000   0.6379550  0.07142857
```

```
## 5 Aurignacien      |||      3  0.0000000  0.4308271  0.33333333
## 6 Aurignacien      |||      4  0.0000000  0.5070802  0.25000000
##   rm.chars
## 1 0.6363636
## 2 1.0000000
## 3 0.7500000
## 4 1.0000000
## 5 1.0000000
## 6 1.0000000
```

Preprocessing

```
# select sign strings of certain length
estimations.df <- estimations.df[estimations.df$num.char > 10,]

# select only certain subcorpora for plotting
estimations.df.selected <- estimations.df[estimations.df$subcorpus == "Aurignacien"|
#estimations.df$subcorpus == "Aurignacien/Gravettian" |
estimations.df$subcorpus == "Cuneiform (Uruk V)"|
#estimations.df$subcorpus == "Cuneiform (Uruk IV)" |
#estimations.df$subcorpus == "Cuneiform (Uruk III)" |
estimations.df$subcorpus == "TeDDi", ]
```

Density Plots

Get subset of data points to plot labels.

```
labels.df <- estimations.df.selected[estimations.df.selected$num.char == 11 &
estimations.df.selected$subcorpus != "TeDDi" &
estimations.df.selected$subcorpus != "Cuneiform (Uruk III)",]
```

Entropy rate vs. unigram entropy

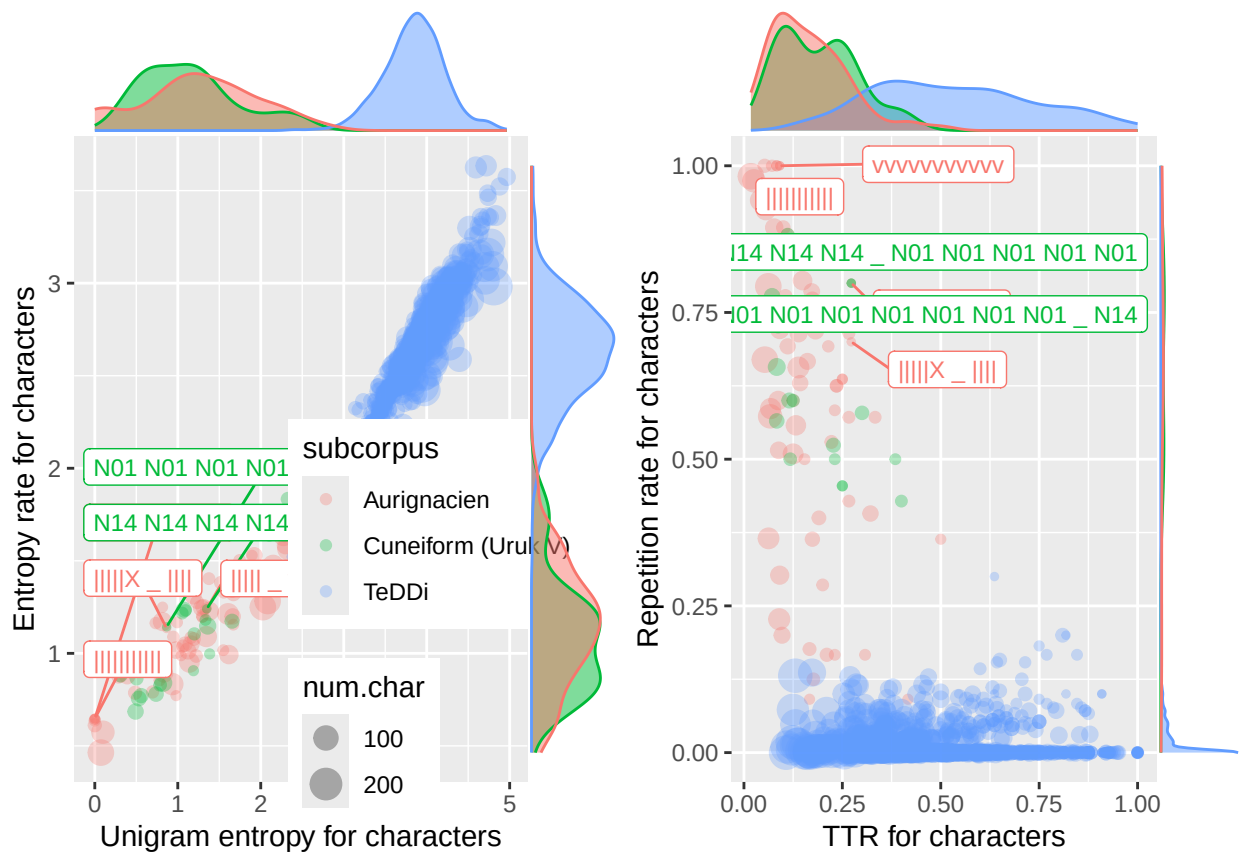
```
huni.hrate.chars.plot <- ggplot(estimations.df.selected, aes(x = huni.chars, y = hrate.chars,
colour = subcorpus, size = num.char)) +
  geom_point(alpha = 0.3) +
#geom_density2d(alpha = .8) +
labs(x = "Unigram entropy for characters", y = "Entropy rate for characters") +
geom_label_repel(data = labels.df,
aes(label = sequence),
position = position_nudge_repel(x = 0, y = 0.5),
size = 3,
max.overlaps = Inf,
show.legend = FALSE) +
theme(legend.position = c(.8, .25))
#facet_wrap(~subcorpus)
huni.hrate.chars.plot <- ggMarginal(huni.hrate.chars.plot, groupFill = T, groupColour = T,
type = "density")
#huni.hrate.chars.plot
```

TTR vs. repetition rate

```
ttr.rm.chars.plot <- ggplot(estimations.df.selected, aes(x = ttr.chars, y = rm.chars,
  colour = subcorpus, size = num.char)) +
  geom_point(alpha = 0.3) +
  theme(legend.position = "none") +
  labs(x = "TTR for characters", y = "Repetition rate for characters") +
  geom_label_repel(data = labels.df,
    aes(label = sequence),
    position = position_nudge_repel(x = 0.25, y = 0),
    size = 3,
    max.overlaps = Inf,
    show.legend = FALSE)
ttr.rm.chars.plot <- ggMarginal(ttr.rm.chars.plot, groupFill = T, groupColour = T,
  type = "density")
#ttr.rm.chars.plot
```

Combine figures.

```
combined.plot <- grid.arrange(huni.hrate.chars.plot, ttr.rm.chars.plot, ncol = 2)
```



Safe complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/combinedPlot_chars.pdf",
  combined.plot, dpi = 300, width = 12, height = 6, device = cairo_pdf)
```